

# Flatcar Container Linux Overview

Flatcar Container Linux is a minimal, immutable, container-optimized Linux OS and the active successor to CoreOS Container Linux (EOL 2020). It is a CNCF project. The name comes from the flat railcar used to transport shipping containers.

## Five Core Tenets

| Tenet                                    | Description                                                                                                                                                       |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Immutable &amp; image-based</b>       | <code>/usr</code> is read-only and dm-verity protected. No package manager. Updates replace all OS binaries atomically including kernel and <code>initrd</code> . |
| <b>Minimal &amp; container-optimized</b> | Ships only Docker, <code>containerd</code> , and basic node tools. Applications run as containers or <code>systemd-sysext</code> images.                          |
| <b>Fully automated</b>                   | Declarative Ignition/Butane config applied once at first boot. No configuration drift.                                                                            |
| <b>Tested &amp; self-updating</b>        | 100+ automated scenario tests. Atomic A/B updates with rollback. No Beta with known issues ever transitions to Stable.                                            |
| <b>Community stewarded</b>               | CNCF project. Multi-employer maintainers. Not vendor driven.                                                                                                      |

## Update Mechanism

Flatcar uses the **USR-A / USR-B** dual-partition scheme (same as ChromeOS). One partition is active, the other is standby. Updates apply to the standby partition; a reboot switches to it. Boot failure automatically falls back to the previous partition.

### Update channels:

| Channel | Cadence   | Support                |
|---------|-----------|------------------------|
| Alpha   | Frequent  | Community              |
| Beta    | ~2 months | Community              |
| Stable  | ~2 months | Community / Pro        |
| LTS     | 12 months | 18 months (commercial) |

## Documentation Map

| Wiki Page                            | Source Section                | Coverage                                      |
|--------------------------------------|-------------------------------|-----------------------------------------------|
| <a href="#">flatcar-provisioning</a> | fb-provision/ , sys-ext/      | Ignition, Butane, CLC, sysext, Terraform      |
| <a href="#">flatcar-deploy</a>       | deploy/                       | Cloud, VMs (KubeVirt, QEMU), bare metal       |
| <a href="#">flatcar-os-config</a>    | os-config/                    | Network, storage, systemd, host config        |
| <a href="#">flatcar-orchestrate</a>  | orchestrate/                  | Docker, Kubernetes, clusters                  |
| <a href="#">flatcar-security</a>     | security/                     | Encryption, hardening, cert auth              |
| <a href="#">flatcar-updates</a>      | updates-releases/             | Nebraska, release channels, update strategies |
| <a href="#">flatcar-diagnostics</a>  | diagnostics/                  | Toolbox, logs, btrfs, rollback                |
| <a href="#">flatcar-devguide</a>     | devguide/ , coreos-migration/ | SDK, supply chain, CoreOS migration           |

## Getting Started

### Quickstart (local QEMU test):

```
# Download image and launch script
wget https://stable.release.flatcar-linux.net/amd64-usr/current/flatcar_production_qemu.sh
chmod +x flatcar_production_qemu.sh
wget https://stable.release.flatcar-linux.net/amd64-usr/current/flatcar_production_qemu_image.img

# Keep a pristine copy for re-provisioning
mv flatcar_production_qemu_image.img flatcar_production_qemu_image.img.fresh
cp --reflink=auto flatcar_production_qemu_image.img.fresh flatcar_production_qemu_image.img

# Boot with a Butane-generated Ignition config
./flatcar_production_qemu.sh -i ignition.json
ssh -p 2222 core@127.0.0.1
```

### Provisioning flow:

```
Butane YAML → butane transpile → Ignition JSON → passed as userdata → applied at first boot
```

# Flatcar vs CoreOS Container Linux

Drop-in replacement. Key naming changes:

| CoreOS                           | Flatcar                           |
|----------------------------------|-----------------------------------|
| <code>coreos.first_boot=1</code> | <code>flatcar.first_boot=1</code> |
| <code>coreos.config.url</code>   | <code>ignition.config.url</code>  |
| <code>coreos.oem.id</code>       | <code>flatcar.oem.id</code>       |
| <code>coreos-installer</code>    | <code>flatcar-installer</code>    |

Both naming conventions are supported in current releases during migration.

## Community

- Slack: `#flatcar` in Kubernetes Slack
- Discord: <https://discord.gg/PMYjFUsJyq>
- GitHub: <https://github.com/flatcar>